

Codebook: Off-Campus Work Intensity, Academic Performance, and Persistence

Synthetic classroom dataset for undergraduate student success analysis (N = 600).

Study Context

This dataset focuses on students who reported off-campus employment during the academic term. The central issue is whether the intensity of off-campus work is related to academic performance and persistence after accounting for other student, financial, academic, institutional, and psychosocial factors. The key threshold is high work intensity, defined as working more than 20 hours per week off campus.

The variable set was designed to reflect the higher education literature on work, engagement, and persistence. Bui emphasizes that students working off campus should be treated as a distinct group and that working more than 20 hours per week was associated with lower persistence. Kuh and colleagues emphasize that student success is multidimensional: grades, persistence, engagement, satisfaction, skill development, and progress toward goals all matter. Nakajima and colleagues highlight the need to analyze multiple factors together when studying persistence, including academic preparation, financial context, work hours, enrollment intensity, English proficiency, faculty interaction, goals, and self-efficacy.

Key Constructs

Off-campus work intensity

The central predictor is `off_campus_work_hours`, the number of hours per week students worked in paid off-campus employment. `high_work_intensity` identifies students working more than 20 hours per week, a threshold often used in research to distinguish moderate work from more intensive work that may compete with academic responsibilities. This construct should be interpreted in context: employment can provide income, independence, career skills, and social connections, but high time demands can reduce study time, rest, and participation in campus life.

Academic performance

Academic performance is represented by `term_gpa`, `cumulative_gpa`, `credits_earned`, and `course_completion_rate`. These indicators capture both grades and progress through attempted coursework. GPA reflects course performance, while credits earned and completion rate reflect whether students are successfully converting enrollment into academic progress. These outcomes may be influenced by prior preparation, engagement, self-efficacy, work demands, financial strain, and institutional support.

Persistence

Persistence is represented by `persistence_score` and `enrolled_next_year`. `persistence_score` is a continuous indicator of academic continuity and intention to remain enrolled, while `enrolled_next_year` is a 0/1 outcome showing whether the student was enrolled the following academic year. Persistence is not only an individual choice; it can reflect academic progress, finances, institutional support, family responsibilities, work schedules, and sense of belonging.

Academic engagement

Academic engagement captures students' investment of time and effort in educationally purposeful activities. Items include studying, attending class, participating, interacting with faculty, and learning with peers. In the student success literature, engagement is important because it represents behaviors and institutional experiences that are more changeable than fixed background characteristics.

Financial strain

Financial strain measures difficulty paying tuition, living costs, food/housing, and the need to work for income. It is included because employment is often connected to financial pressure, but the relationship is not simple: some students work because they must cover basic costs, while others work for independence, career exploration, or spending money. Financial strain can confound the relationship between work hours and academic outcomes.

Institutional support and belonging

Institutional support reflects advising quality, perceived faculty care, belonging, and use of resources. These variables are included because persistence is shaped not only by individual motivation but also by whether students experience the college as responsive and navigable. Supportive conditions may help students manage competing responsibilities.

Academic self-efficacy

Self-efficacy refers to students' confidence in their ability to manage time, complete assignments, ask for help, and recover from setbacks. It is related to academic behavior because students who believe they can manage coursework may be more likely to persist through difficulty and use available supports.

Work-academic conflict

Work-academic conflict captures the degree to which paid work interferes with studying, class schedules, energy in class, and campus participation. This construct helps distinguish the number of work hours from the lived consequences of those hours. Two students may work the same number of hours but differ substantially in schedule control, exhaustion, and interference with coursework.

Career value of work

Work career value measures whether employment helps students apply classroom learning, build career skills, develop professional networks, and stay motivated. This construct recognizes that employment is not only a risk factor. Some work experiences may support students' goals, especially when the job is related to the major or future career.

Mental strain and exhaustion

Mental strain captures stress, exhaustion, sleep problems, and feeling overwhelmed. It is included because competing academic, work, and financial responsibilities can affect wellbeing, and wellbeing can in turn shape academic performance and persistence.

Variable Dictionary

Likert items use a 1-5 response scale where higher values indicate more of the named experience. Binary variables use 1 = yes/present and 0 = no/absent unless otherwise noted.

Variable	Type	Range / Units	Meaning
participant_id	integer	1-600	Unique participant identifier.
age	integer	years	Student age at the time of the survey.
age_20_or_above	integer	0/1	Indicator for whether the student was age 20 or older. <i>1 = age 20+, 0 = age 18-19</i>
gender	factor	category	Self-reported gender. <i>woman; man; nonbinary_or_other</i>
race_ethnicity	factor	category	Self-identified race/ethnicity. <i>white; hispanic_latino; black; asian; multiracial_or_other</i>
first_generation	integer	0/1	Whether the student is a first-generation college student. <i>1 = first-generation; 0 = continuing-generation</i>
family_income_bracket	ordered factor	low < middle < high	Approximate family income category. <i>low; middle; high</i>
pell_grant	integer	0/1	Whether the student received Pell Grant aid. <i>1 = yes; 0 = no</i>
loan_status	integer	0/1	Whether the student used student loans during the academic year. <i>1 = yes; 0 = no</i>
dependent_care	integer	0/1	Whether the student had regular caregiving responsibilities for a child, adult, or family member. <i>1 = yes; 0 = no</i>
institution_control	factor	category	Institution control/type. <i>public; private_nonprofit; private_for_profit</i>
institution_selectivity	ordered factor	open_access < minimally_selective < moderately_selective < highly_selective	Broad selectivity category of the institution. <i>open_access; minimally_selective; moderately_selective; highly_selective</i>
tuition_fees_thousand	numeric	thousands of dollars	Approximate annual tuition and fee level in thousands of dollars.
major_group	factor	category	Broad field of study. <i>stem; business; health; social_science; arts_humanities; other</i>
commute_minutes	integer	minutes	Typical one-way commute time to campus.
hs_gpa	numeric	0.00-4.00	High school GPA before college.
english_proficiency	numeric	1.00-5.00	Self-rated English academic reading/writing proficiency. <i>higher = more proficient</i>
remedial_coursework	integer	0/1	Whether the student was placed into at least one developmental/remedial course. <i>1 = yes; 0 = no</i>
enrollment_status	factor	category	Current term enrollment intensity. <i>full_time; part_time</i>
enrollment_units	numeric	6.0-18.0 credits/units	Number of credits/units attempted in the current term.
credits_attempted	numeric	credits/units	Credits attempted in the current term.
credits_earned	numeric	credits/units	Credits successfully completed in the current term.
course_completion_rate	numeric	0.00-1.00	Proportion of attempted credits completed successfully.
off_campus_work_hours	integer	hours/week	Hours worked per week in off-campus employment.
high_work_intensity	integer	0/1	High-intensity off-campus work indicator based on the >20 hours/week threshold used in the Bui study. <i>1 = more than 20 hours/week; 0 = 20 or fewer hours/week</i>

Variable	Type	Range / Units	Meaning
oncampus_work_hours	integer	hours/week	Hours worked per week in on-campus employment.
total_work_hours	integer	hours/week	Total weekly paid work hours from on-campus and off-campus jobs.
job_related_to_major	integer	0/1	Whether the student reported that the off-campus job was related to their academic major or career plans. <i>1 = yes; 0 = no</i>
schedule_variability	integer	1-5	Perceived unpredictability of work schedule. <i>higher = more variable</i>
schedule_control	integer	1-5	Perceived control over work schedule. <i>higher = more control</i>
study_hours_per_week	numeric	hours/week	Typical hours per week spent studying outside class.
sleep_hours	numeric	hours/night	Typical nightly sleep duration.

Variable	Type	Range / Units	Meaning
fin_strain_pay_tuition	integer	1-5 Likert item	Item contributing to the financial strain construct. <i>higher = more of the named experience</i>
fin_strain_living_costs	integer	1-5 Likert item	Item contributing to the financial strain construct. <i>higher = more of the named experience</i>
fin_strain_food_housing	integer	1-5 Likert item	Item contributing to the financial strain construct. <i>higher = more of the named experience</i>
fin_strain_need_work_income	integer	1-5 Likert item	Item contributing to the financial strain construct. <i>higher = more of the named experience</i>
engage_study_effort	integer	1-5 Likert item	Item contributing to the academic engagement construct. <i>higher = more of the named experience</i>
engage_class_attendance	integer	1-5 Likert item	Item contributing to the academic engagement construct. <i>higher = more of the named experience</i>
engage_class_participation	integer	1-5 Likert item	Item contributing to the academic engagement construct. <i>higher = more of the named experience</i>
engage_faculty_contact	integer	1-5 Likert item	Item contributing to the academic engagement construct. <i>higher = more of the named experience</i>
engage_peer_learning	integer	1-5 Likert item	Item contributing to the academic engagement construct. <i>higher = more of the named experience</i>
support_advising_quality	integer	1-5 Likert item	Item contributing to the institution support construct. <i>higher = more of the named experience</i>
support_faculty_care	integer	1-5 Likert item	Item contributing to the institution support construct. <i>higher = more of the named experience</i>
support_belonging	integer	1-5 Likert item	Item contributing to the institution support construct. <i>higher = more of the named experience</i>
support_resource_use	integer	1-5 Likert item	Item contributing to the institution support construct. <i>higher = more of the named experience</i>
efficacy_manage_time	integer	1-5 Likert item	Item contributing to the self efficacy construct. <i>higher = more of the named experience</i>
efficacy_complete_assignments	integer	1-5 Likert item	Item contributing to the self efficacy construct. <i>higher = more of the named experience</i>
efficacy_ask_for_help	integer	1-5 Likert item	Item contributing to the self efficacy construct. <i>higher = more of the named experience</i>
efficacy_recover_setbacks	integer	1-5 Likert item	Item contributing to the self efficacy construct. <i>higher = more of the named experience</i>
work_conflict_missed_study	integer	1-5 Likert item	Item contributing to the work academic conflict construct. <i>higher = more of the named experience</i>
work_conflict_schedule_overlap	integer	1-5 Likert item	Item contributing to the work academic conflict construct. <i>higher = more of the named experience</i>
work_conflict_tired_in_class	integer	1-5 Likert item	Item contributing to the work academic conflict construct. <i>higher = more of the named experience</i>
work_conflict_missed_campus	integer	1-5 Likert item	Item contributing to the work academic conflict construct. <i>higher = more of the named experience</i>
work_value_applies_learning	integer	1-5 Likert item	Item contributing to the work career value construct. <i>higher = more of the named experience</i>
work_value_career_skills	integer	1-5 Likert item	Item contributing to the work career value construct. <i>higher = more of the named experience</i>
work_value_professional_network	integer	1-5 Likert item	Item contributing to the work career value construct. <i>higher = more of the named experience</i>
work_value_motivation	integer	1-5 Likert item	Item contributing to the work career value construct. <i>higher = more of the named experience</i>
wellbeing_stress	integer	1-5 Likert item	Item contributing to the mental strain construct. <i>higher = more of the named experience</i>

Variable	Type	Range / Units	Meaning
wellbeing_exhaustion	integer	1-5 Likert item	Item contributing to the mental strain construct. <i>higher = more of the named experience</i>
wellbeing_sleep_problems	integer	1-5 Likert item	Item contributing to the mental strain construct. <i>higher = more of the named experience</i>
wellbeing_overwhelmed	integer	1-5 Likert item	Item contributing to the mental strain construct. <i>higher = more of the named experience</i>
financial_strain_index	numeric	see meaning	Mean of 4 Likert items measuring difficulty paying for college and living costs. <i>higher = more of construct</i>
academic_engagement_index	numeric	see meaning	Mean of 5 Likert items measuring academic effort, participation, faculty contact, and peer learning. <i>higher = more of construct</i>
institution_support_index	numeric	see meaning	Mean of 4 Likert items measuring advising quality, faculty care, belonging, and resource use. <i>higher = more of construct</i>

Variable	Type	Range / Units	Meaning
self_efficacy_index	numeric	see meaning	Mean of 4 Likert items measuring confidence in managing academic tasks and setbacks. <i>higher = more of construct</i>
work_academic_conflict_index	numeric	see meaning	Mean of 4 Likert items measuring the extent to which work interferes with study, class, and campus participation. <i>higher = more of construct</i>
work_career_value_index	numeric	see meaning	Mean of 4 Likert items measuring perceived career and learning value from work. <i>higher = more of construct</i>
mental_strain_index	numeric	see meaning	Mean of 4 Likert items measuring stress, exhaustion, sleep problems, and feeling overwhelmed. <i>higher = more of construct</i>
term_gpa	numeric	see meaning	Current term GPA.
cumulative_gpa	numeric	see meaning	Cumulative college GPA.
persistence_score	numeric	see meaning	Continuous persistence-related score based on intention, perceived likelihood, and continuity of enrollment. <i>higher = more of construct</i>
enrolled_next_year	integer	see meaning	Observed/recorded enrollment persistence outcome for the following academic year.
not_persisted	integer	see meaning	Inverse of enrolled_next_year.

Source Notes

The dataset is conceptually grounded in research on off-campus work intensity and persistence among college students, literature on student engagement and student success, and prior work on community college persistence. These sources motivate the inclusion of academic, financial, engagement, and psychosocial controls.

Bui, B. C. (2025). Striving for College Success: The Persistence of Students who Work Off-Campus. *Innovative Higher Education*, 50, 1945-1968.

Kuh, G. D., Kinzie, J., Buckley, J. A., Bridges, B. K., & Hayek, J. C. (2006). What Matters to Student Success: A Review of the Literature.

Nakajima, M. A., Dembo, M. H., & Mossler, R. (2012). Student Persistence in Community Colleges. *Community College Journal of Research and Practice*, 36(8), 591-613.